

Atum TRON Contracts Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

Dacian

Al-qaqa

August 27, 2025

Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Atum Labs on Atum TRON Contracts. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated August 27, 2025, was delivered to Atum Labs, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Atum Labs or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Atum TRON Contracts code provided by Atum Labs. The findings consist of 1 Critical & 2 Low severity issues with the remainder being informational.

Summary

Project Name	Atum TRON Contracts
Repository	tvm-contracts
Commit	e6d5e9f79552...
Repository 2	permit2-tron
Commit	a31ae2274278...
Audit Timeline	Aug 18th - Aug 20th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	2	1	1
Informational	3	2	1
Gas Optimizations	0	0	0
Total	6	4	2

6 Audit Scope

Cyfrin has previously audited Atum's EVM Contracts. The scope of this audit was Atum's TRON contracts and the fork of the Permit2 library for TRON deployment:

```
tvm-contracts/contracts/Escrow.sol
tvm-contracts/contracts/FulfillmentProxy.sol
tvm-contracts/contracts/TIP712.sol
permit2-tron/tron.diff
```

In addition to the Permit2 diff, we also read the remaining unchanged Permit2 code.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Atum is a cross-chain stablecoin payments protocol that leverages an off-chain Request for Quote (RFQ) mechanism to facilitate token transfers across different blockchains and stablecoin types. The protocol's design abstracts away blockchain complexity from the end users (Originators), who simply sign a Permit2 message to initiate a payment request; they never need to interact directly with smart contracts or manage cross-chain messaging.

When an Originator needs to pay in stablecoins across chains (e.g., deposit USDC on Ethereum but pay using DAI on Arbitrum), they publish an RFQ containing their signed Permit2 authorization, which multiple liquidity providers (Settlers) can bid on competitively. The winning Settler locks the source funds in the `Escrow` contract and fulfills the swap on the destination chain, after which the fulfillment is verified and the escrowed funds are released to the Settler.

This architecture pushes the technical complexity, including liquidity management, cross-chain execution and blockchain interactions, to Settlers who compete to provide the best rates, while Originators get a simple, signature-based experience. It is suited to applications like cross-border payments, remittances and chain-agnostic stablecoin transfers.

9 Attestation

Cyfrin attests that it performed a security review of the Atum TRON Contracts code provided by Atum Labs.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-08-27-cyfrin-atum-tron-contracts-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Atum Labs has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=a30625654ff5631ee492b8075bc7c42e5ff72798ad4ae203f2d4e16a3d816ab4
F=2025-08-27-cyfrin-atum-tron-contracts-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.